

Differentially Private Directed Networks

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October 2024

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0.1 Differentially Private Directed Networks

Source: Yan [2021]

This paper has many similarities with Karwa and Slavković [2016], however instead of assuming a simple undirected network, they propose a more involved method for directed networks.

0.1.1 Contributions of the paper

1. **Noisy sequence** Similar to the Karwa and Slavković [2016], they propose using discrete laplace mechanism.
2. **Estimator of p_0 model** Estimating with differentially private bi-degree sequence is consistent and asymptotically normal as in Karwa and Slavković [2016]
3. **Additional asymptotic variance term for large noise**
4. **Denoising bi-degree sequence as a $l - 1$ minimization as in Karwa and Slavković [2016]**

0.1.2 Statistical Inference

Definition 1 (Directed β networks) Let $A = a_{i,j}$ be the adjacency matrix where $a_{i,j}$ is indicator variable for an edge from node i to node j .

$$a_{i,j} = \begin{cases} 1 & \text{if there is a directed edge from } i \text{ to } j \\ 0 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

Also, define

$$d_i^+ = \sum_{j \neq i} a_{i,j} \quad (\text{out-degree of node } i)$$

$$d_i^* = \sum_{j \neq i} a_{j,i} \quad (\text{in-degree of node } i)$$

Proposition 1 (Noise mechanism for bi-degree sequence) Using the Algorithm ?? from Karwa and Slavković [2016], add two independent random noise from discrete Laplace distribution with parameter $\exp(-\epsilon_n/2)$ to both in-degree and out-degree sequences. Then resulting noisy bi-degree sequence is differentially private.

Proof 1 (Noise mechanism for bi-degree sequence) Proof follows exactly same as in Karwa and Slavković [2016].

p_0 model: This is a model defining the distribution of the graph using in and out degree sequences. Consider G_n as the graph, then probability of obtaining this graph in this model is

$$\mathbb{P}(G_n) = \frac{1}{c(\alpha, \beta)} \exp \left(\sum_i \alpha_i d_i^+ + \sum_j \beta_j d_j^- \right)$$

where $\alpha = (\alpha_1, \dots, \alpha_n)^T$, $\beta = (\beta_1, \dots, \beta_n)^T$ and $c(\alpha, \beta) = \sum_{j \neq i} \log(1 + \exp(\alpha_i + \beta_j))$

Also, we have given probability under the same model

$$\mathbb{P}(a_{i,j} = 1) = \frac{\exp(\alpha_i + \beta_j)}{1 + \exp(\alpha_i + \beta_j)}$$

Note that this model assumes that $\beta_n = 0$ for identifiability.

With the same approach as we mentioned before for [Karwa and Slavković \[2016\]](#), likelihood equations for bi-degree sequence can be defined as follow.

$$\begin{aligned} z_i^+ &= \sum_{j \neq i} \frac{e^{\alpha_i + \beta_j}}{1 + e^{\alpha_i + \beta_j}}, \quad i = 1, \dots, n \\ z_j^- &= \sum_{i \neq j} \frac{e^{\alpha_i + \beta_j}}{1 + e^{\alpha_i + \beta_j}}, \quad j = 1, \dots, n-1 \end{aligned}$$

where z_i^+ and z_j^- are noisy versions of d_i^+ and d_j^- respectively. Let's define $\theta = (\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_n)^T$. Then $\hat{\theta}$, solution for above equations, is the differentially private estimator of θ . In the following sections, we are going to discuss asymptotic properties of this estimator.

0.1.3 Asymptotic properties of the estimator

One important remark for this section is that number of parameters increase with number of nodes, therefore classical statistical techniques to guarantee asymptotic consistency is not valid in this case. Therefore, a version of Newton method is applied to estimate parameter θ .

Now, define

$$\begin{aligned} F_i(\theta) &= z_i^+ - \sum_{k=1, k \neq i}^n \frac{e^{\alpha_k + \beta_j}}{1 + e^{\alpha_k + \beta_j}}, \quad i = 1, \dots, n, \\ F_{n+j}(\theta) &= z_j^- - \sum_{k=1, k \neq i}^n \frac{e^{\alpha_k + \beta_j}}{1 + e^{\alpha_k + \beta_j}}, \quad j = 1, \dots, n \end{aligned}$$

and note that solution of $F(\theta) = 0$, say $\hat{\theta}$, is the estimator. Proposed Newton step for finding $\hat{\theta}$ is

$$\theta^{(k+1)} = \theta^{(k)} - [F'(\theta^{(k)})]^{-1} F(\theta^{(k)})$$

Therefore, using the convergence rate for iterative sequence, we can state the following theorem.

Theorem 1 (Uniform Consistency) *Given the true parameter θ^* , if $\epsilon^{-1}e^{12\|\theta^*\|_\infty} = o((n \log n)^{1/2})$, then $\hat{\theta}$ exists and satisfies*

$$\|\theta^* - \hat{\theta}\|_\infty = O_p\left(\frac{1}{\epsilon_n} \frac{(\log n)^{1/2} e^{6\|\theta^*\|_\infty}}{n^{1/2}}\right) = o_p(1)$$

Also, if $\hat{\theta}$ exists, it is unique.

Definition 2 (Sub-exponential family) *A random variable is called sub-exponential with parameter κ if*

$$(\mathbb{E}|X|^p)^{1/p} \leq \kappa p, \quad \text{for all } p \geq 1$$

Definition 3 (Composition inequality for sub-exponential) *Let $X_1, \dots, X_n$ be independent centered random variables, and suppose each X_i is sub-exponential with parameter κ_i . Define $\kappa = \max_i \kappa_i$. Then we have*

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_i X_i\right| \geq \epsilon\right) \leq 2 \exp\left[\gamma n \cdot \min\left(\frac{\epsilon}{\kappa^2}, \frac{\epsilon}{\kappa}\right)\right]$$

Lemma 1 (Laplace distribution is sub-exponential) *Let X be continuous Laplace random variable with $f(x) = (2\lambda)^{-1}e^{-|x|/\lambda}$, then X is also sub-exponential random variable with $\kappa = \lambda$. Moreover, if X is a discrete Laplace with pmf $\frac{1-\lambda}{1+\lambda}\lambda^{|x|}$, then X is a sub-exponential with $\kappa = 2(\log \frac{1}{\lambda})^{-1}$.*

Definition 4 ($\mathcal{L}(n, M)$ - a specific matrix family) *If $V \in \mathcal{L}(n, M)$ with $M/m = o(n)$, then for sufficiently large n*

$$\|V^{-1} - S\| \leq \frac{c_1 M^2}{m^3(n-1)^2}$$

for constant c_1 . Here S can be used to estimate inverse of V if V is in this specific family.

Theorem 2 (Convergence of Newton iterative sequence) *Consider*

$$\begin{aligned} F_i(\theta) &= d_i - \sum_{k=1, k \neq i}^n f(\alpha_i + \beta_k), \quad i = 1, \dots, n \\ F_{n+j}(\theta) &= b_j - \sum_{k=1, k \neq j}^n \quad j = 1, \dots, n-1 \\ F(\theta) &= (F_1(\theta), \dots, F_n(\theta), F_{n+1}(\theta), \dots, F_{2n-1}(\theta))^T \end{aligned}$$

where $f(\cdot)$ has third derivative. Given $D \subset \mathbb{R}^{2n-1}$, assumptions of the theorem are as follows.

1. *For any $x, y, v \in D$, we have*

$$\begin{aligned} \|[F'(x) - F'(y)]v\|_\infty &\leq K_1 \|x - y\|_\infty \|v\|_\infty \\ \max_{i=1, \dots, 2n-1} \|F'_i(x) - F'_i(y)\|_\infty &\leq K_2 \|x - y\|_\infty \end{aligned}$$

2. $\|F(\theta^*)\| = O_p(\sqrt{n \log n})$
3. $\theta^{(0)} \in D$ with $\Omega(\theta^{(0)}, 2r) \subset D$ where $r = \|[F'(\theta^{(0)})]^{-1} F(\theta^{(0)})\|_\infty$
4. Assume $F'(\theta) \in \mathcal{L}_n(m, M)$ or $-F'(\theta) \in \mathcal{L}_n(m, M)$
5. We have $\rho r \rightarrow 0$ as $n \rightarrow \infty$, where ρ is defined as

$$\rho = \frac{c_1(2n-1)M^2K_1}{2m^3n^2} + \frac{K_2}{(n-1)m}.$$

Note that $F'(\theta)$ is the Jacobian matrix of F on θ , and $F'_i(\theta)$ is the gradient function of F on θ .

Then, under the assumptions 1-5, $\theta^{(k)} \in \Omega(\theta^{(0)}, 2r)$ for $k = 1, 2, \dots$ are well defined and satisfy

$$\|\theta^{(k+1)} - \theta^{(0)}\|_\infty \leq \frac{r}{(1 - \rho r)}$$

Additionally, $\lim_{k \rightarrow \infty}(\theta^{(k)})$ is the solution for $F(\theta) = 0$.

Proof 2 (Uniform Consistency) Given the above theorems, lemmas and definitions, we can prove the uniform consistency of the estimator derived under privacy constraints. To do this, we need to check the assumptions of the newton convergence theorem. If all the assumptions are satisfied, this means that limit of the sequence will be our estimator and it will be uniformly consistent with the true parameter if we start the sequence from the true value θ^* .

1. This is satisfied with $K_1 = n - 1$ and $K_2 = (n - 1)/2$
2. We need to show that $\|F(\theta^*)\| = O_p(\sqrt{n \log n})$. This is true because

$$\max \left(\max_i |z_i^+ - \mathbb{E}(d_i^+)|, \max_i |z_i^- - \mathbb{E}(d_i^-)| \right) = O_p(\sqrt{n \log n} + \kappa_n \sqrt{\log n})$$

where $\kappa_n = 2(-\log \lambda_n)^{-1} = 4/\epsilon_n$. This equality comes from the concentration inequality of discrete Laplace distribution and property of directed networks.

3. $\theta^{(0)}$ is chosen as θ^* and $\theta \in \Omega(\theta^*, 2r)$ with $r = \|[F'(\theta^*)]^{-1} F(\theta^*)\|_\infty$
4. If $\theta \in \Omega(\theta^*, 2r)$ then $-F'(\theta) \in \mathcal{L}_n(m, M)$ with certain m and M values.
5. Following the 2nd item, $r \leq O(\frac{(\log n)^{1/2} \epsilon_n^{-1} e^{6\|\theta^*\|_\infty}}{n^{1/2}})$ ($O_p?$), so $r = o(1)$ ($o_p?$) when $(1 + \kappa_n)e^{6\|\theta^*\|_\infty} = o((n/\log n)^{1/2})$. Also, following the 1st item, $\rho = O(e^{6\|\theta^*\|_\infty})$. Then if $(1 + \kappa_n)e^{6\|\theta^*\|_\infty} = o((n/\log n)^{1/2})$, $\rho r \rightarrow 0$ as $n \rightarrow \infty$.

Since all the assumptions are satisfied, $\lim(\theta^k)$ exists, and

$$\|\hat{\theta} - \theta^*\|_\infty = O\left(\frac{(\log n)^{1/2} \epsilon_n^{-1} e^{6\|\theta^*\|_\infty}}{n^{1/2}}\right) = o(1)$$

($O_p, o_p?$)

In Meijia's paper, consistency was proven using the following assumptions of Newton's method.

Theorem 3 (Convergence of Newton iterative sequence v2) *Let D be an open convex set of R^n and $F : D \rightarrow R^n$ be Frechet differentiable with a Jacobian $F'(\cdot)$. Suppose there exists $x_0 \in D$, constants $L_0, \eta > 0$. Under the following assumptions*

1. $\{F'(x_0)\}^{-1}$ is bounded linear operator,
2. $\|\{F'(x_0)\}^{-1}F(x_0)\| \leq \eta$
3. $\|\{F'(x_0)\}^{-1}(F'(x) - F'(x_0))\| \leq L_0\|x - x_0\|$, for all $x \in D$
4. $10L_0\eta \leq 1$
5. $\bar{B}(x_0, 2\eta) \in D$

sequence generated by Newton iteration is well-defined and remains in $\bar{B}(x_0, s^)$ for some $0 < s^* < 2\eta$, also we have*

$$\|x^{(t)} - x_0\| \leq \frac{\eta}{2^{t-1}}$$

The estimator is not only consistent, but for any fixed k first k element of $(\hat{\theta} - \theta^*)$ has asymptotically normal distribution. For this purpose, we need Fisher information matrix due to the asymptotic properties of MLE estimator, and more importantly, we need inverse of Fisher information matrix. For this purpose, first define $g = (d_1^+, \dots, d_n^+, d_1^-, \dots, d_n^-)$ and $\tilde{g} = (z_1^+, \dots, z_n^+, z_1^-, \dots, z_n^-)$. By Taylor's expansion to each component of $\tilde{g} - \mathbb{E}g$, we obtain second-order term as $V(\hat{\theta} - \theta)$ where V is Fisher information matrix. Since the Fisher information belongs to certain family of matrices we can approximate it with a specific invertible matrix, say S . Therefore, proving that $S(\tilde{g} - \mathbb{E}g)$ is approximately normal is enough.

Theorem 4 (Asymptotic Normality) *If $\epsilon_n^{-1}(\log n)^{1/2}e^{2\|\theta^*\|_\infty} = o(1)$, then for any fixed $k \geq 1$ as $n \rightarrow \infty$, the vector consisting first k elements of $(\hat{\theta} - \theta^*)$ is asymptotically multivariate normal with mean 0 and covariance matrix given by the upper-left $k \times k$ block of S , approximation for Fisher information matrix.*

Bibliography

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